

Assess learning and plan next steps

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Order 0.72, $\frac{7}{10}$ and 73%. Copy or complete the first step.

2. Calculate $\frac{5}{6} + \frac{7}{12}$.

3. Find 37.5% of 640.

4. Miro says: Miro chooses a familiar method before identifying the number type and relationship. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Order 0.72, $\frac{7}{10}$ and 73%. Copy or complete the first step.
Answer $\frac{7}{10}$, 0.72, 73%.
2. Calculate $\frac{5}{6} + \frac{7}{12}$.
Answer $1\frac{5}{12}$.
3. Find 37.5% of 640.
Answer 240.
4. Miro says: Miro chooses a familiar method before identifying the number type and relationship. Is this correct? Explain.
Answer Name the domain first, represent the relationship, then choose and check the calculation.

Assess learning and plan next steps

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Order 0.72, $\frac{7}{10}$ and 73%.

6. Check this result using an inverse, estimate or known fact:
Find 37.5% of 640.

2. Calculate $\frac{5}{6} + \frac{7}{12}$.

7. Prove or explain your reasoning: Simplify 45:60 and find an equivalent ratio with first part 12.

3. Find 37.5% of 640.

8. Identify and correct this misconception: Miro chooses a familiar method before identifying the number type and relationship.

4. Solve this independently and show every stage: Order 0.72, $\frac{7}{10}$ and 73%.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Calculate $\frac{5}{6} + \frac{7}{12}$.

10. Write and solve a similar question to: Find 37.5% of 640.

Teacher answers and checking prompts

1. Order 0.72, $\frac{7}{10}$ and 73%.
Answer $\frac{7}{10}$, 0.72, 73%.
2. Calculate $\frac{5}{6} + \frac{7}{12}$.
Answer $1\frac{5}{12}$.
3. Find 37.5% of 640.
Answer 240.
4. Solve this independently and show every stage: Order 0.72, $\frac{7}{10}$ and 73%.
Answer $\frac{7}{10}$, 0.72, 73%.
5. Use a second method or representation for: Calculate $\frac{5}{6} + \frac{7}{12}$.
Answer Expected result: $1\frac{5}{12}$. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Find 37.5% of 640.
Answer Expected result: 240. A valid check must be shown.
7. Prove or explain your reasoning: Simplify 45:60 and find an equivalent ratio with first part 12.
Answer 3:4 and 12:16. The explanation must show why the method works.
8. Identify and correct this misconception: Miro chooses a familiar method before identifying the number type and relationship.
Answer Name the domain first, represent the relationship, then choose and check the calculation.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Find 37.5% of 640.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Assess learning and plan next steps

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Simplify 45:60 and find an equivalent ratio with first part 12.

6. Create a new example that tests the same idea as: Find 37.5% of 640.

2. Prove the answer to this question, rather than only calculating it: Order 0.72, $\frac{7}{10}$ and 73%.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Calculate $\frac{5}{6} + \frac{7}{12}$.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 240.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro chooses a familiar method before identifying the number type and relationship.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Simplify 45:60 and find an equivalent ratio with first part 12.
Answer 3:4 and 12:16. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Order 0.72, $\frac{7}{10}$ and 73%.
Answer Expected result: $\frac{7}{10}$, 0.72, 73%. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Calculate $\frac{5}{6} + \frac{7}{12}$.
Answer Expected result: $1\frac{5}{12}$. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 240.
Answer One valid reconstruction is: Find 37.5% of 640. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro chooses a familiar method before identifying the number type and relationship.
Answer Name the domain first, represent the relationship, then choose and check the calculation.
6. Create a new example that tests the same idea as: Find 37.5% of 640.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.